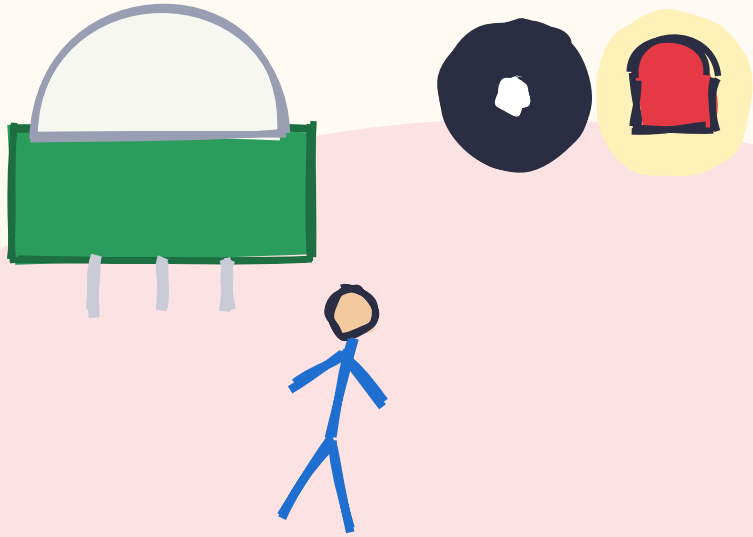
Little Explorers · Class 8

Motion Detector
Room Alarm

Can you sneak past our robot guard?

Motion sensor **Beep!** **Find the blind spot**



Did you know?
Your body is warm, about 37°C. Special sensors can feel that warmth even across a room!

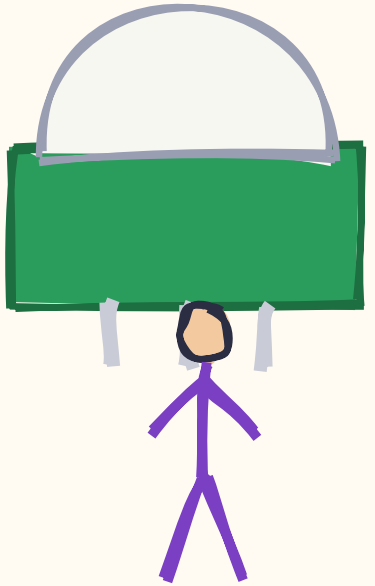
LEARN

Let's wonder...

How does a light know you walked in?

Lights that switch on by themselves

Some lights in shops and toilets turn on **by themselves** when you walk in. Nobody touched a switch! How did the light know you were there?



A camera?

Magic?

A motion sensor!

Today: we build an **alarm** that beeps when someone moves!

Learning objectives

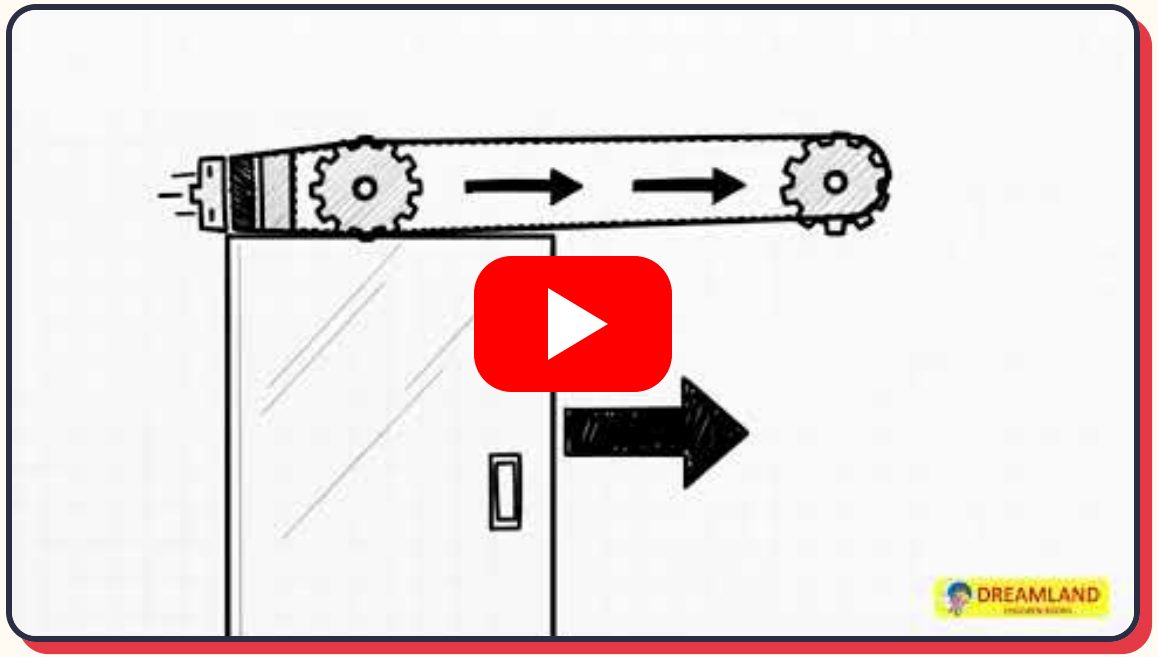
Today I can...

1	**Body heat:** say that our bodies give off **warmth** a sensor can feel	**Proof:** feel the warmth of my hands and explain it
2	**Alarm:** make the alarm **beep** when someone moves	**Proof:** our alarm beeps when I wave
3	**Blind spot:** find where the alarm **cannot see**	**Proof:** we marked the blind spot on our map

LEARN

Objective 2 · Alarm · video

Video time!



How an Automatic Door Works

Dreamland Publications · about 2 minutes · youtu.be/xCX2y3zOgBA

Watch for...

1. What makes the door **open**?
2. Where is the sensor on the door?
3. Where have you seen a door like this?

Scan to watch

After the video: Walk towards your partner slowly. They open their arms like a door when you get close!

Objective 1 · Body heat

Our bodies glow with warmth

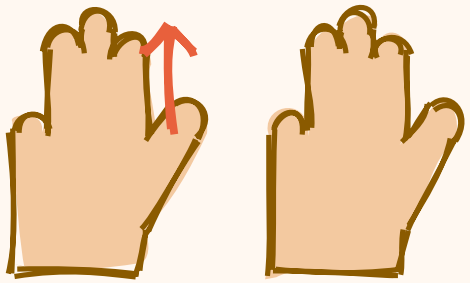
The idea

Every person and animal gives off **warmth**. We cannot see it, but special sensors can **feel** it.

Our motion sensor looks for warmth that **moves**. A warm person walking past? **Got you!**

Freeze!
If you stand perfectly still, the sensor may not notice you!

Example: warm hands

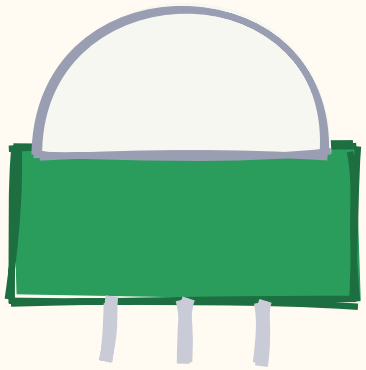


Rub your hands together fast. Now hold them near your cheek. Can you **feel** the warmth without touching? The sensor feels people the same way!

LEARN

Objective 1 · Body heat · meet our helper

The motion sensor



The white dome

The bumpy white dome is like a **bug's eye**. It looks in lots of directions at once.

Fun fact

Bees and flies have eyes made of many tiny parts too!

Give it a minute

When we switch it on, it needs about **1 minute** to wake up. Be patient!

Real world

Where are motion sensors?



Lights
Switch on in corridors and toilets



Alarms
Keep homes and shops safe at night



Automatic doors
Open when shoppers walk up



Taps and dryers
Hand dryers start when you hold out your hands



Where have you seen something switch on all by itself?

Saving energy
Lights that switch off in empty rooms save electricity.

LEARN

Our machine

Feel, think, show



How it works

The **sensor** feels warm movement.
The Arduino **thinks**: did something move?
Yes? **Beep!** and the red light flashes.

Teacher: parts are pre-wired on the breadboard. Children plug in the USB cable, snap the picture blocks in mBlock and test.

Objective 3 · Blind spot

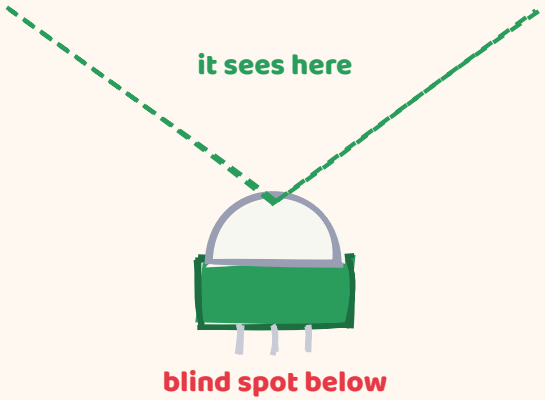
Every guard has a blind spot

The idea

Our sensor can only see **in front** of it, in a wide fan shape. Behind it, under it and far away are **blind spots**.

Engineers test their alarms to find blind spots, then move the sensor to a better place.

Example: a car mirror



A driver cannot see everything in the mirrors. The places they cannot see are called **blind spots**. That is why we never stand behind a car!

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Sleeping guard
2. Snap the picture blocks
3. Can you sneak past?
4. Alarm challenges

MAKE

Class game

Sleeping guard

How to play

1. One child is the **guard** and closes their eyes.
2. Everyone else creeps slowly towards the guard.
3. The guard listens and points: "I heard you move!"
4. That person freezes and goes back to the start.
5. Who can touch the guard's shoulder first?

Then discuss

Was it easier to catch people who moved **fast** or **slow**? Our sensor is the same!



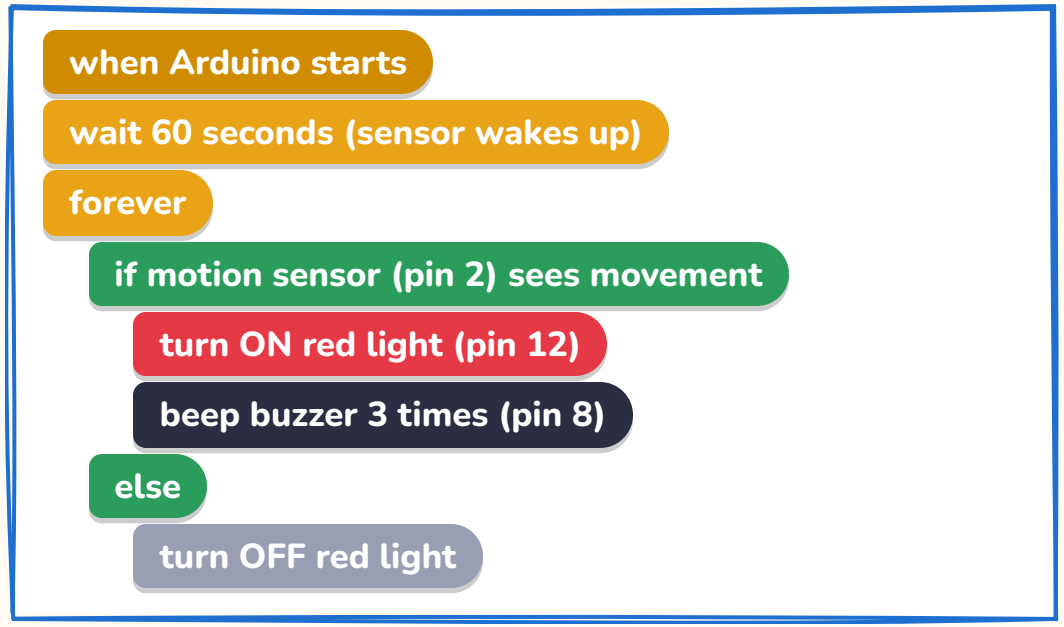
Big idea

Alarms notice MOVEMENT. Moving slowly or staying still can fool them.

MAKE

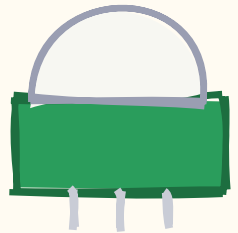
Time to build!

Snap the picture blocks



Read it like a story

Wait for the sensor to wake up.
IF something moves: light and beep!
ELSE: stay quiet.
Keep watching forever.



Too noisy?
Change 'beep 3 times' to 1 time. Quieter for the classroom!

MAKE

Sneak test!

Can you sneak past?

Try this	I think…	What happened
Walk normally in front	beep / quiet?	_____
Creep very slowly	beep / quiet?	_____
Crawl under it	beep / quiet?	_____
Walk behind it	beep / quiet?	_____

Talk about it

Where is the **blind spot**?

Where is the **best place** to put the sensor in our classroom?

Draw a map

Draw the classroom. Colour where the sensor sees in green and the blind spots in red.

MAKE

Level up!

Alarm challenges

Choose your challenge



Siren: make the buzzer go high-low-high-low like a police car.



Welcome light: turn on a light that says hello instead of an alarm.



Count visitors: count how many times the sensor beeps in 5 minutes.



Best spot: move the sensor to remove the blind spot.

Think like an engineer

Should the alarm beep when the class pet hamster moves? How could we stop that?

Boss level

Make a 'shh' button that turns the alarm off!

Objective check

Can I really do it? Prove it.

01	Why can the sensor feel a person? (Hint: warmth)	○ ○ ○
02	Show your alarm beeping when you wave.	○ ○ ○
03	Point to a blind spot on your map.	○ ○ ○

Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Let each group try to sneak past another group's alarm.

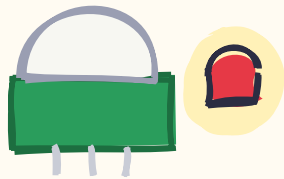
Show what you know

What did we learn?

- [x] Our bodies give off warmth
- [x] The motion sensor feels warm things moving
- [x] The alarm beeps when someone moves
- [x] Every sensor has blind spots

At home

Look for motion sensors in shops or at school. Where are they?



Exit ticket: how could you sneak past a motion alarm?

Next class: build a Waving Robot Arm!